Programming for Computational Biology (R)
Assignment I

Total marks: 20

Dataset provided

You are provided with three CSV files:

- abundance.csv: Species-level read counts for 60 stool samples.
- metadata.csv: Sample information, including disease status, age, gender and location.
- taxonomy.csv: Taxonomic information corresponding to the species in the abundance table.

The dataset contains 30 colorectal cancer (CRC) samples and 30 healthy samples, with 308 bacterial species.

Instructions

1. For all random simulations, use set.seed(123) immediately before generating random values to ensure reproducibility.
2. Submit a complete, commented R script (.R) and a short report (.pdf) containing numerical results, relevant tables and brief statistical and biological interpretations.

Part A. Species prevalence and abundance (5 marks)

A1. Identify the five most prevalent bacterial species across all 60 samples. Calculate their prevalence as the number and percentage of samples in which each species is detected (read count greater than zero). In case of ties, rank species alphabetically by species name. **(2 marks)**

A2. Calculate the relative abundance (%) of the five selected species in every sample. Use the total read count across all 308 species in the corresponding sample as the denominator. Report the mean and median relative abundance of each species across all 60 samples. **(2 marks)**

A3. Does the most prevalent species also have the highest median relative abundance? Support your answer using the results from A1 and A2. **(1 mark)**

Part B. Descriptive statistics and group comparison (5 marks)

Use the five species selected in Part A.

B1. Calculate the mean, median and interquartile range (IQR) of relative abundance for each species separately in CRC and healthy samples. Present the results in a table. Report all values as relative abundance percentages. **(2 marks)**

B2. Which of the five species shows the largest absolute difference in median relative abundance between CRC and healthy samples? Calculate the difference in percentage points and identify the group with higher median relative abundance. **(2 marks)**

B3. Which species and disease group show the greatest standard deviation (SD) in relative abundance? Interpret what this indicates about variation between samples. **(1 mark)**

Part C. Probability and independence (5 marks)
Use the sample metadata.

C1. Calculate the probabilities of randomly selecting an individual who belongs to the CRC group, is female, and belongs to both categories (CRC and female). Report all three probabilities. **(1 mark)**

C2. Are disease status and gender independent in this cohort? Compare the observed joint probability of CRC and female with the product of their marginal probabilities. Support your conclusion using a two-way table of observed and expected counts, where expected counts are calculated assuming independence. **(2 marks)**

C3. Assume that the probability of CRC is estimated from the provided dataset. For a randomly selected group of 10 individuals, calculate the probability of obtaining exactly 4 CRC individuals and at most 4 CRC individuals using dbinom() and pbinom(), respectively. Simulate 1,000 such groups using rbinom() and compare the simulated probability of exactly 4 CRC individuals with the theoretical probability. **(2 marks)**

For this calculation, assume that individuals are selected independently with the estimated probability of CRC.

Part D. Normal distribution and simulation (5 marks)
Use the age variable in the metadata.

D1. Calculate the mean and standard deviation of age. Generate 1,000 simulated age values using rnorm(), with the observed mean and standard deviation as the parameters. Compare the simulated mean and standard deviation with those of the observed data. **(1 mark)**

D2. Using the fitted normal distribution, calculate the density at age 50 using dnorm() and the probability of being aged 50 years or younger using pnorm(). Explain the difference between these two quantities. **(1 mark)**

D3. Calculate the theoretical probability of being between 40 and 60 years old using pnorm(). Compare it with the observed proportion of individuals in the dataset whose ages fall within this interval. **(1 mark)**

D4. Assess whether the observed age data follow an approximately normal distribution using the Shapiro–Wilk normality test (shapiro.test()). Report the test statistic and p-value, and interpret the result using a significance level of 0.05. **(2 marks)**